

ALEJANDRO ITO ARAMENDIA

London, UK | alejandro.itoaramendia@gmail.com | LinkedIn: [linkedin.com/in/alejandro5902](https://www.linkedin.com/in/alejandro5902)

Personal portfolio: alejandro-ito-aramendia.netlify.app | GitHub: github.com/Alex5902 | 07598807434

EDUCATION

Royal Holloway University – MSc Machine Learning	Sep 2025 – Sep 2026
Tohoku University, Japan – BEng Mechanical and Aerospace Engineering	Oct 2021 – Sep 2025
Relevant Coursework: Linear Algebra 1 (A), Linear Algebra 2 (A), Probability and Statistics (A), Data Structures and Algorithms (AA)	
King's College London Maths School, UK – A-Levels	Sep 2019 – Aug 2021
A*AAA in Maths, Further Maths, Physics and Computer Science (AS)	

EXPERIENCE

Research Assistant (Software Developer) – Ishii Lab, Tohoku University	Oct 2023 – Sep 2024
- Developed a MATLAB GUI to automate complex vector fluid flow analysis, increasing data processing efficiency for research teams and enhancing multi-dataset analysis by over 300%. - Applied bilinear interpolation and regression models to analyse 10+ tissue flow sequences, enabling more precise speed/direction mapping for clinical ultrasound research.	
Business Strategy Intern – KGT JIGE Internship, Kansai University	Sep 2024 – Nov 2024
- Led a team in developing a detailed expansion strategy for a sustainable start-up targeting the German automotive market, running competitor analysis and supply chain risk across 3 entry models. - Selected as one of the top 3 out of 13 teams to present to 100+ attendees, demonstrating the ability to synthesise research into actionable strategy under real-world constraints.	

PROJECTS

Data-Free LoRA Adapter Merging for Medical AI (Thesis Project)
- Developed data-free LoRA merging algorithms (Factor-Aware Merging) in Python, using SVD and PCA for subspace projection to align 5 distinct adapters and mitigate task interference. - Designed a novel two-stage residual correction framework that isolates and corrects task-specific model errors by denoising residual matrices with truncated SVD before applying a TIES-merge. - Set a new state-of-the-art accuracy of 57.41% on a 5-modality medical VQA benchmark (+7.25% over baseline), successfully consolidating 61.2% of specialist knowledge into a single mode. - Managed large-scale experiments on a Linux-based HPC cluster, using Slurm for job scheduling and Bash scripts to automate model training, evaluation, and hyperparameter sweeps.
Weather Driven Electricity Demand Causal Analysis (Python)
- Designed a causal inference workflow combining Bayesian Structural Time-Series (BSTS) and PCMCI over 1,440 hourly data points (60 days), identifying lagged dependencies in GB electricity demand. - Engineered 100+ time-series features (lags, cyclic encodings), developed ARIMA and Prophet baselines, and visualised 24-hour periodicity autocorrelations using libraries tigramite, causalimpact, and Prophet. - Estimated a +14,000 MWh uplift (+£840,000 wholesale cost, +294 tons CO₂ emissions) during the 2025 heatwave, demonstrating useable insights for grid forecasting, risk management, and sustainability reporting.
Technical Writing on Machine Learning and Algorithms (2000+ Likes)
Authored 10+ in-depth technical guides on deep learning and algorithmic design, gaining 2000+ likes and praise from PhDs for my clarity and depth.

SKILLS

Languages/Frameworks: Python, JavaScript, SQL, MATLAB, pandas, numpy, pytorch, scikit-learn, seaborn

Machine Learning/Data Science: Data Visualisation, Supervised Learning, Error/Uncertainty Analysis, Causal Inference, CNNs, VLMs, Test-Time-Adaptation, PEFT, Feature Engineering, LoRA

Databases: PostgreSQL, MongoDB, Redis

Cloud/DevOps: AWS (Lambda, API Gateway, RDS, S3), Docker, Git, CI/CD Pipelines, Linux, Bash, HPC & Slurm Schedulers, Postman, Agile

Systems/Tools: RESTful APIs, GPU Compute (Bash-scripted), JWT, Jest (Unit Testing)

Languages: English (Native), Spanish (Advanced), Japanese (JLPT N2)